

RESEARCH NOTEBOOK

Why does multilingual RAG miss answers in another language?

It reads Hindi, German and English. It can still skip the best answer because of the language it's in.

SUBJECT *dense retrieval · RAG*

METHOD *SHIFT · EMNLP '26 Findings*

REPLICATION *Qdrant · mES-small · XRAG*

GERMAN QUERY

„Wie funktioniert ein Transformer?“

GERMAN PASSAGE

So-so answer
same language as the query

ranked #1

ENGLISH PASSAGE

Best answer
right meaning, other language

✗ not in Top-K

Soubhagya Jain
open the notes ↪



1 What I assumed

An embedding turns text into coordinates. Similar meaning should mean nearby points.

In a multilingual model, I expected the language to drop out completely.

the textbook view

one topic, one cluster

● EN "Reset my password"

● HI "मेरा पासवर्ड रीसेट करें"

● DE "Mein Passwort zurücksetzen"

*Meaning decides position.
Language shouldn't.*

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2 What can happen

For some models, a more honest picture of a vector is:

$$\text{vector} \approx \text{meaning} + \text{language} + \text{model geometry}$$

0.61

relevant answer in the
query's language, in top
10 (✓)

0.08

equally relevant answer
in another language (✓)

one model, one
dataset

Recall@10 · multilingual-e5-small · XRAG,
15k news articles in 5 languages

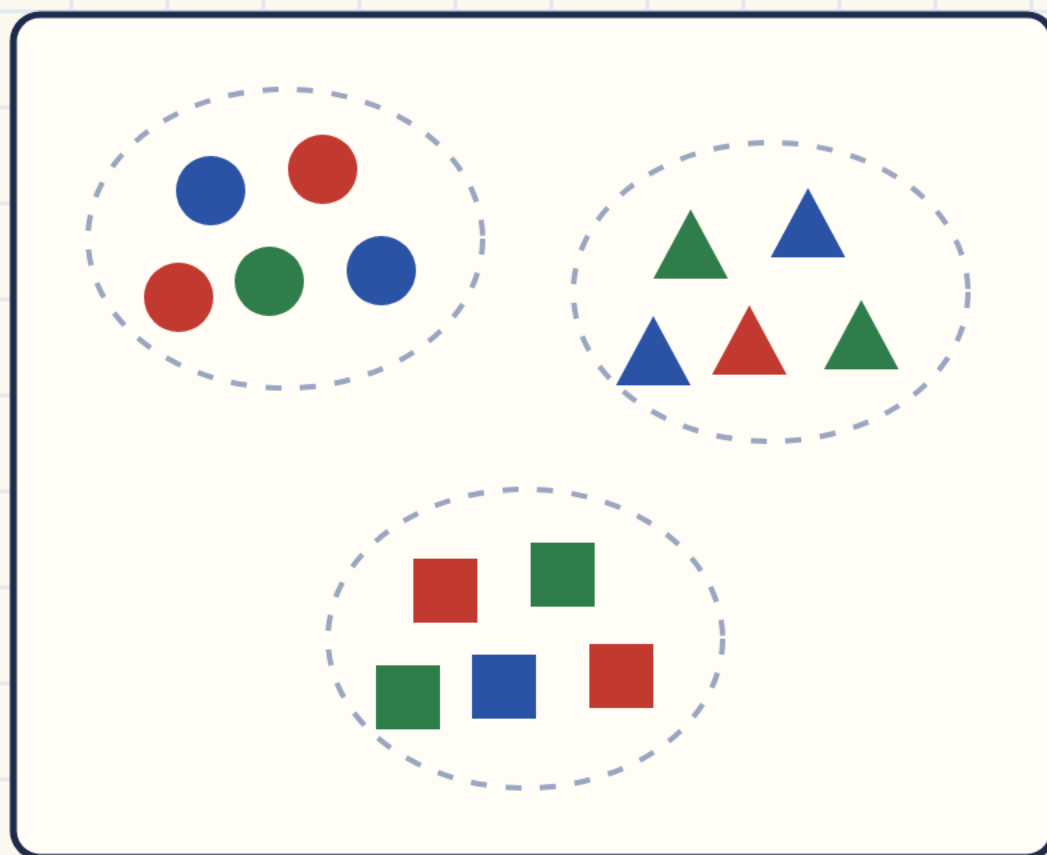
Same relevance. Very different recall.

turn the page ↪

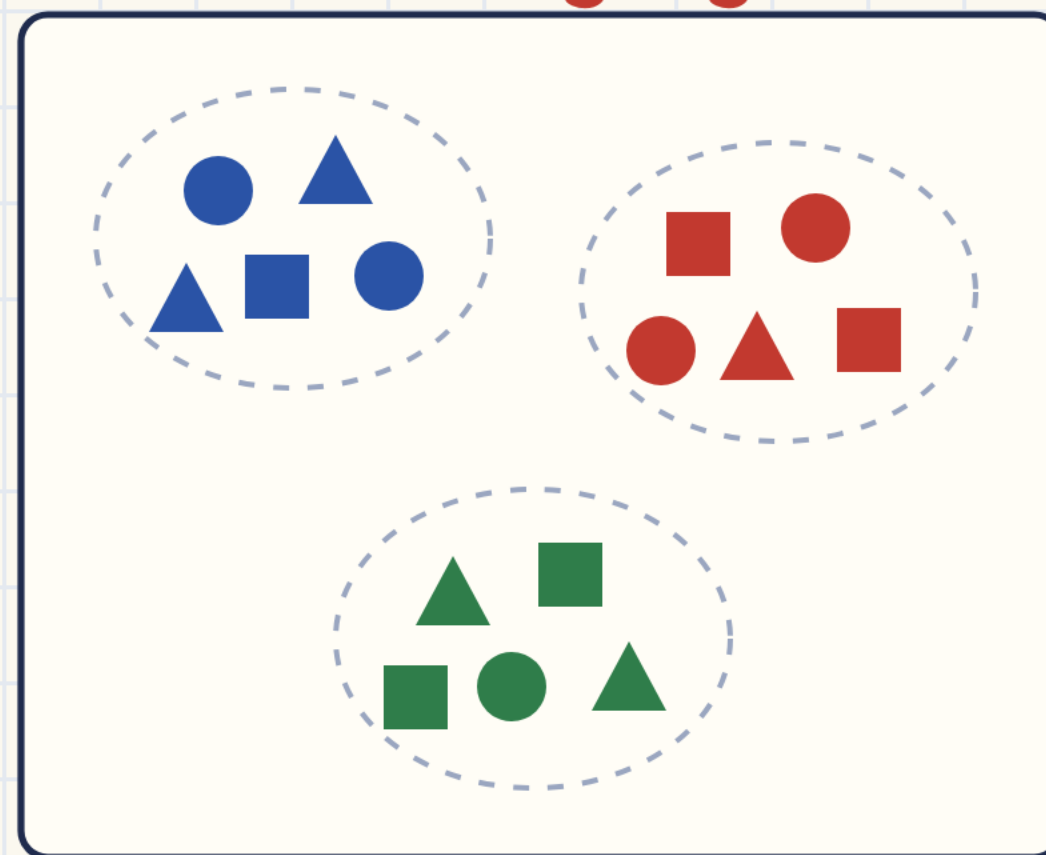
3 Colour vs shape

Plot the vectors in 2D. Colour = language. Shape = topic.

Ideal



Warning sign



● EN ● HI ● DE ● ▲ ■ = three topics

Colours clumping across topics means language may be steering position.

*Illustration.
UMAP/PCA is a
hint, not proof.*

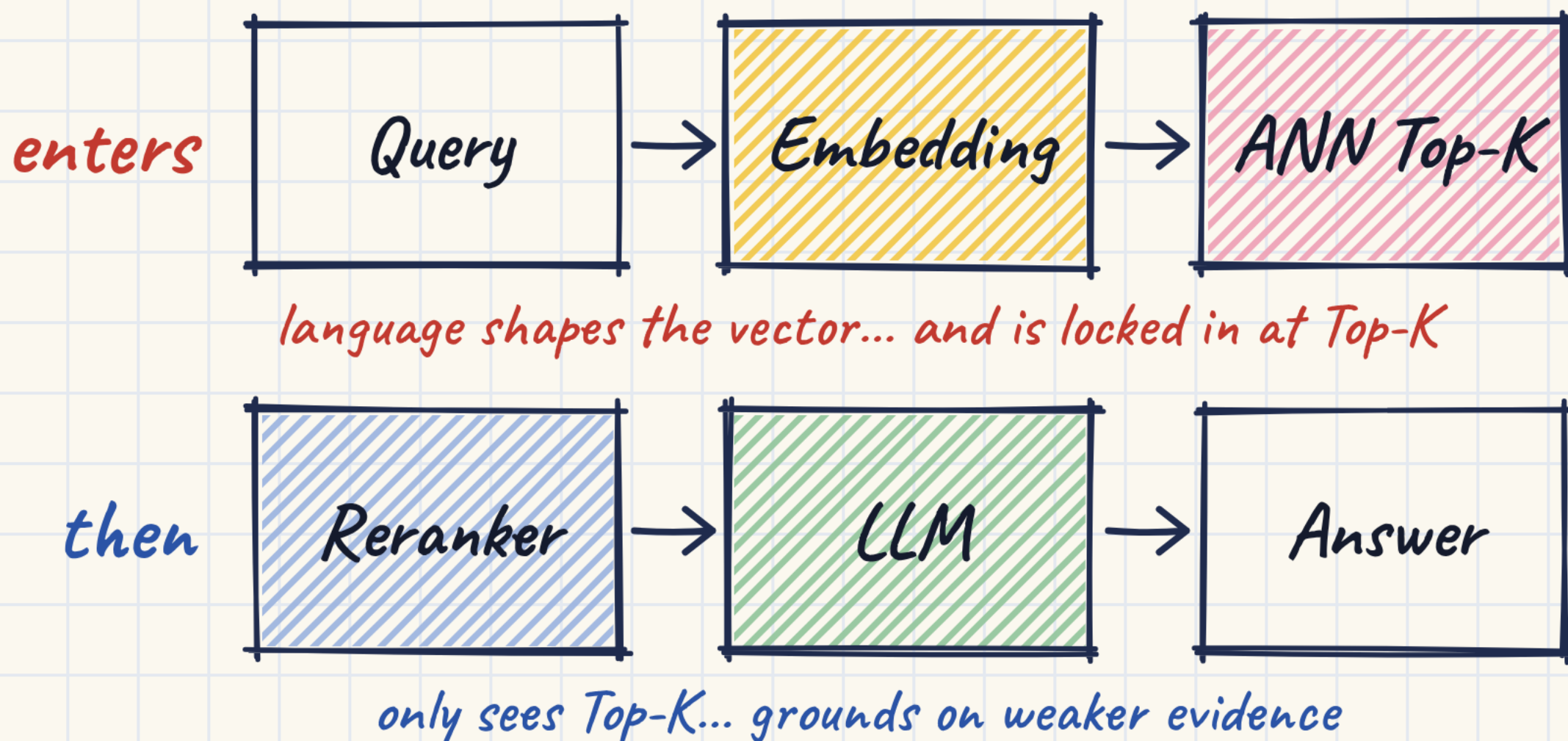
The retrieval metrics are the evidence.

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4

Why RAG breaks

The bias enters early, and everything downstream inherits it.



The index isn't the culprit. Qdrant found HNSW matched exact search within ~ 0.005 ✓

The LLM can only ground on what Top-K lets through.

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5 Rerankers can't rescue

ONE SPANISH QUERY · QDRANT · XRAG

English answer at rank #438 (✓)

Say your reranker scores the top 50

So it never sees the best answer

Shifted same article reached rank #5 (✓)

Recall@20 low →
candidate problem.
Fix retrieval first.

Recall@20 fine,
MRR low →
ranking problem.
A reranker helps.

Candidate recall ≠ ranking.

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6 The SHIFT idea

Training-free, applied at index time (Jang et al., 2026).

THE RECIPE

<i>Pairs</i>	a few thousand translations, DE ↔ EN
<i>Offset</i>	$\text{mean}(\text{embed}(\text{DE}) - \text{embed}(\text{EN}))$
<i>Index</i>	each DE vector $- \alpha \cdot \text{offset}$
<i>Query</i>	shift it too if it isn't English

Meaning mostly cancels inside each pair. What's left estimates a language direction.



An intuition, not a clean linear decomposition. Effect varies by model.

nDCG@20, before → after
mE5-large 0.633 → 0.737 ✓
bge-m3 0.715 → 0.717 ✓

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7 My experiment

HINDI QUERY

ट्रांसफॉर्मर कैसे काम करता है?

how does a transformer work?

RELEVANT · ENGLISH

How transformers work

attention, step by step

VS

WEAKER · HINDI

ट्रांसफॉर्मर का इतिहास

a short history, on-topic but not the answer

Which one ranks first?

Repeat across 200 queries. Count how often the weaker Hindi passage wins.

← my own diagnostic,
not a standard
metric

Report Recall@5/10/20, MRR and nDCG@10, always split into same-language vs cross-language.

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8 The trade-off

0.08 → 0.24

cross-language
Recall@10 (V)

0.61 → 0.49

same-language
Recall@10 (V)

Qdrant · multilingual-e5-small · XRAG · after SHIFT

(A) Query and answer usually
share a language
bias may be harmless

(B) Many query languages,
mostly-English corpus
invariance is critical

(C) Localized copies of the same
corpus
*keep some language
awareness*

(D) Must answer in the user's
language
*make language an explicit
rule*

Where does the answer live?

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9 The principle

Before you pick the embedding model, measure:

Where your queries live, language by language.



Where your evidence lives.

Whether language became a ranking feature
nobody chose.



*"A multilingual model is not a
language-neutral retriever by default."*

SOURCES (✓) Jang, Hong, Moon & Lim, "SHIFT", arXiv 2606.18801, EMNLP 2026 Findings ·
(✓) Qdrant, "SHIFTing Languages in Multilingual RAG", Sep 16 2026